

Materials

Playing It Safe With ORGANIC COMPOUNDS



Saurabh Durgopal
is working as
technology
journalist at EFY

Development of integrated flexible electronics is important for the construction of advanced wearable devices. And when you consider our fascination with wearables, demand for newer materials to design such wearables also rises. Such developments in the fields of material technology are happening all around us. Ranging from energy generation to lighting, we have some fascinating materials being tested out in labs. 3D printing materials and organic light emitting diodes (OLEDs) are also among areas of research.

Carbon based materials, for example, are being used to improve efficiency. Perovskites are one such wide-ranging class of materials consisting of organic molecules. These are made mostly of carbon and hydrogen, bonded with a metal and a halogen, in a three-dimensional crystal lattice. These can be made much more economically compared to conventional materials and with fewer emissions.

Solar researches in perovskites

Perovskite based solar cells consist of a transparent conductive oxide-coated glass

substrate. With a top encapsulation, these have an evaporated metal cathode providing high-performance solutions.

On the other hand, use of lead makes these not so safe for use. An alternative is to use tin based perovskites, which are non-poisonous. However, current power-conversion efficiency of tin based devices is significantly lower than of lead based devices. Nevertheless, if their development history is any indication, we can look forward to some non-poisonous solar cells with higher efficiency.

A mix of organic and inorganic ions with crystal structures similar to calcium-titanium-oxide (CaTiO_3) results in hybrids. Perovskites containing halide ions such as fluoride or chlorine are a sub-set of such hybrids. For converting sunlight to electricity, however, iodide variants are a much better option. Materials such as methylammonium-lead-iodide ($\text{CH}_3\text{NH}_3\text{PbI}_3$) can convert sunlight to electricity. These have enabled urban poor communities in India to access off-grid electricity. After all, there is only so much coal and hydropower can do.

Oxford Photovoltaics, UK, has begun commercialising thin-film perovskite solar cells. These are directly printed on silicon solar cells, copper-indium-gallium-selenide (CIGS) solar cells or glass. It is one of the three mainstream thin-film technologies—the other two being cadmium-telluride and amorphous-silicon, with a lab-efficiency above 20 per cent.

Making inexpensive solar panels

Solar energy harvesting is an area that shows a lot of promise in renewable energy. The present panels employ silicon solar cells, which are highly efficient, converting up to 21.5 per cent of sunlight that falls on these into electricity, and are very durable.

However, processing silicon wafers is very expensive. Add to that the required thickness of 0.3mm in order to absorb

Fig. 1: Perovskite fabricated on a glass sheet
(Image courtesy:
physicsworld.com)

